

A Heat Exchanger Simulation

Simulating a shell-and-tube heat exchanger for industrial applications

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TUM Uhrenturm

- 1** Introduction & Goal
- 2 Governing Equations
- 3 Numerical Methods
- 4 Experimental Setup: Water - Water
- 5 An Industrial Application: Molten Salt - Thermal Oil

Introduction

Heat exchangers in solar plants **enhance efficiency** by transferring heat from solar collectors to working fluids, facilitating effective **energy conversion** and **storage**.



Figure 1 Heat exchangers at the *Planta Solar de Almería*¹

¹J. Lappalainen et al. "Modelling a molten salt thermal energy system - A validation study". In: *Applied Energy* (2019).

Goal

Simulate and analyse different scenarios for a shell-and-tube heat exchanger (e.g. with and without baffles, different fluids)

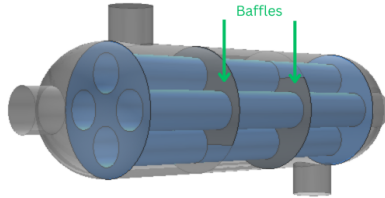


Figure 2 Shell-and-tube HEX geometry²

²SIMFLOW Technologies. *SimFlow*. Version 5.0. URL: <https://sim-flow.com/>.

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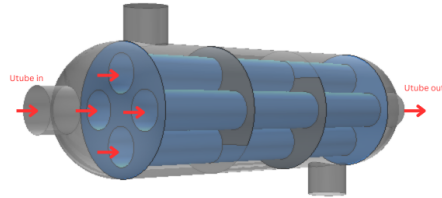


Figure 3 Shell-and-tube HEX geometry³

³SIMFLOW Technologies. *SimFlow*. Version 5.0. URL: <https://sim-flow.com/>.

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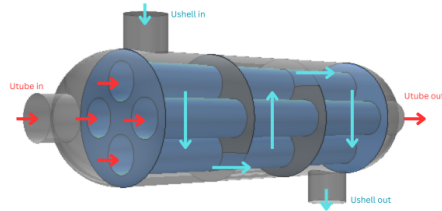


Figure 4 Shell-and-tube HEX geometry⁴

⁴SIMFLOW Technologies. *SimFlow*. Version 5.0. URL: <https://sim-flow.com/>.

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Governing Equations

Transient Setup

Continuity equation:

$$\nabla \cdot u = 0 \quad (1)$$

Momentum equation:

$$\rho \left(\frac{\partial u}{\partial t} + \nabla \cdot (uu) \right) = -\nabla p + \mu \nabla^2 u \quad (2)$$

Energy equilibrium equation:

$$\frac{\partial T}{\partial t} + \nabla \cdot (uT) - \frac{\kappa_f}{\rho_f c_{p,f}} \nabla^2 T = 0 \quad (3)$$

with $u \equiv u(x, t)$, $T \equiv T(x, t)$ and $p \equiv p(x, t)$ for $x \in \Omega \subset \mathbb{R}^3$, $t \in [0, 900]$ ⁵

⁵Christopher Greenshields and Henry Weller. *Notes on Computational Fluid Dynamics: General Principles*. 2022.

Governing Equations & Temperature Coupling

Transient Setup

Governing Equations:

$$\begin{aligned}\nabla \cdot u &= 0 \\ \rho \left(\frac{\partial u}{\partial t} + \nabla \cdot (uu) \right) &= -\nabla p + \mu \nabla^2 u \\ \frac{\partial T}{\partial t} + \nabla \cdot (uT) - \frac{\kappa_f}{\rho_f c_{p,f}} \nabla^2 T &= 0\end{aligned}$$

Coupled Temperature:

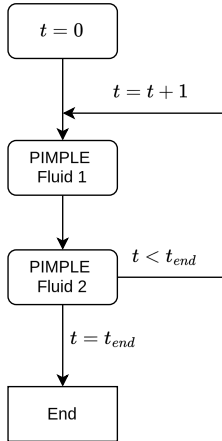
$$\kappa_f \frac{\partial T_{f1}}{\partial n} = \kappa_s \frac{\partial T_{f2}}{\partial n} \quad (4)$$

with $u \equiv u(x, t)$, $T_{-,f1,f2} \equiv T_{-,f1,f2}(x, t)$ and $p \equiv p(x, t)$ for $x \in \Omega \subset \mathbb{R}^3$, $t \in [0, 900]$

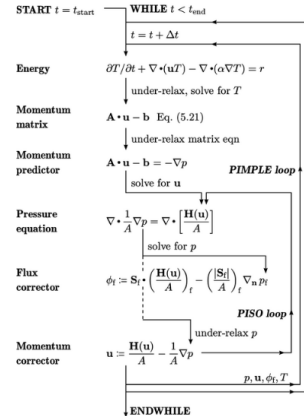
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Overview - *chtMultiRegionFoam*



(a) Overview of *chtMultiRegionFoam*



(b) PIMPLE loop diagram

Discretization	Scheme
Time	Implicit-Euler
Convection	Upwind
Gradients	Gauss Linear
Interpolation	Linear

Variable	Solver	Residual
Pressure	GAMG	10^{-4}
Velocity	PBiCG Stabilized	10^{-4}
Enthalpy	PBiCG Stabilized	10^{-4}
Density	PCG	10^{-4}

Outline

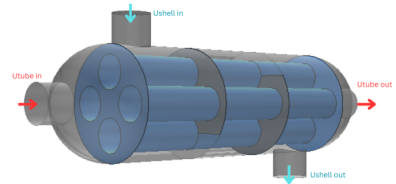
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Boundary Conditions

Water - Water

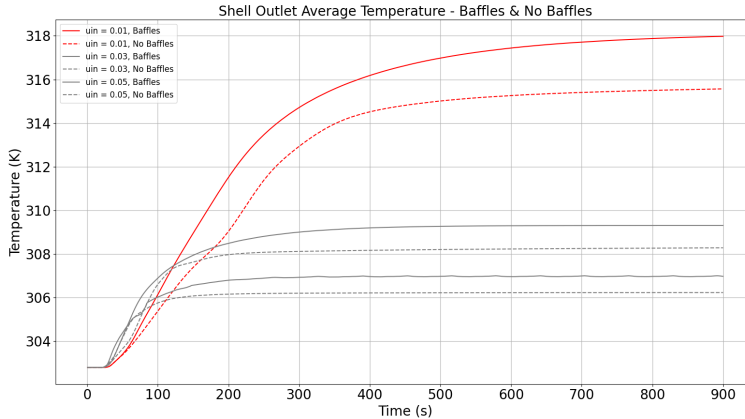
Boundary	Velocity	Temperature
Tube outlet	$\frac{\partial u}{\partial n} = 0$	$\frac{\partial T}{\partial n} = 0$
Tube inner wall	$u_{patch} = 0$	Coupled
Tube outer wall	$u_{patch} = 0$	Coupled
Shell outlet	$\frac{\partial u}{\partial n} = 0$	$\frac{\partial T}{\partial n} = 0$
Shell wall	$u_{patch} = 0$	$\frac{\partial T}{\partial n} = 0$
Tube inlet	$u = 0.1 \text{ m/s}$	$T = 353.15K$
Shell inlet	$u = u_{in} \text{ m/s}$	$T = 302.8K$

Table 1 Boundary Conditions



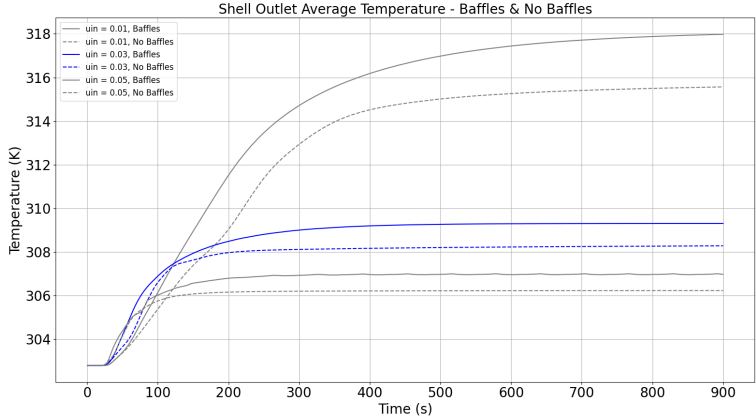
Results

Water - Water



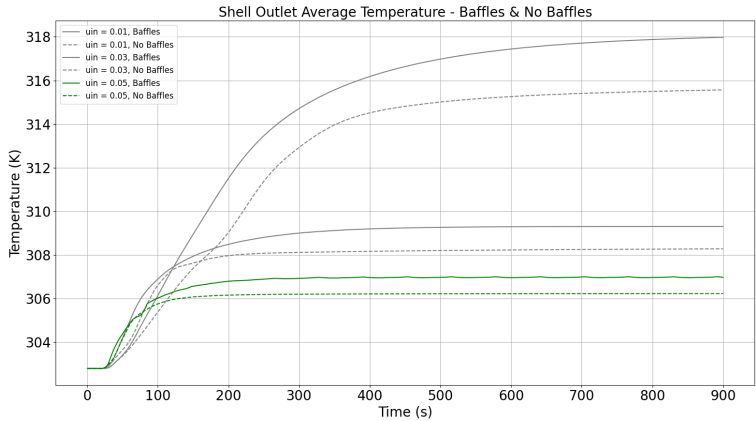
Results

Water - Water



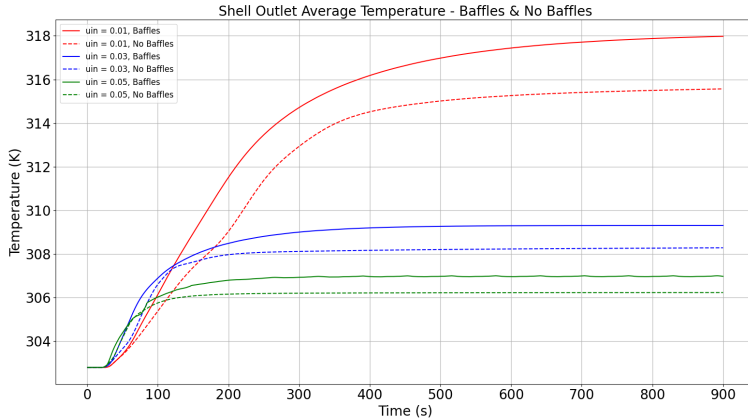
Results

Water - Water



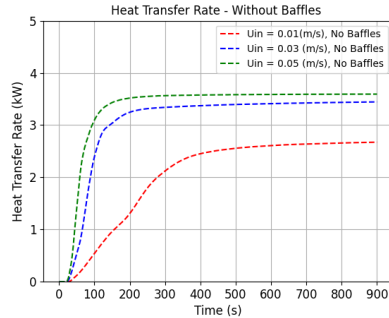
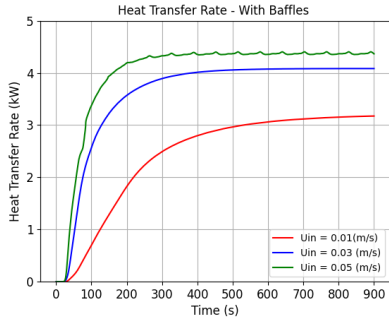
Results

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Water - Water



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Results

Molten Salt - Thermal Oil

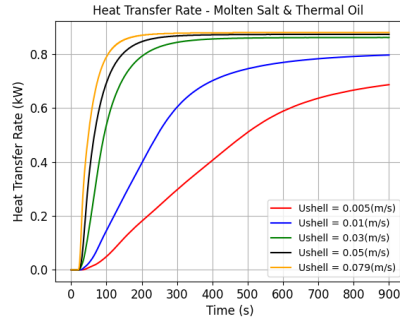
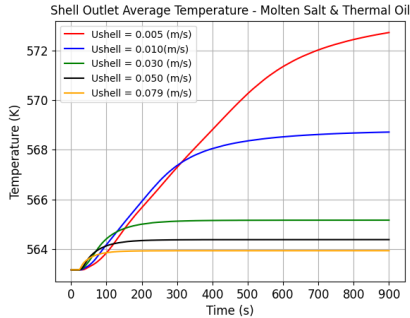


Figure 7 Results for $T_{shell\ in} = 563.15K$, $T_{tube\ in} = 646.15K$

References

-  Greenshields, Christopher and Henry Weller. *Notes on Computational Fluid Dynamics: General Principles*. 2022.
-  Lappalainen, J. et al. “Modelling a molten salt thermal energy system - A validation study”. In: *Applied Energy* (2019).
-  Technologies, SIMFLOW. *SimFlow*. Version 5.0. URL: <https://sim-flow.com/>.